Olist Data Cleaning Log

Live formulas — figures update automatically if the source tables change.

1. Table profile — row counts and key uniqueness

Table	Rows	Key column	Distinct keys	Duplicate rows	Notes
Order_Items_Dataset_FactTable	112,650	order_id + order_item_id	112,650	0	Fact table. order_id repeats by design.
Orders_Dataset	99,441	order_id	99,441	0	Must be unique to act as dimension.
Customer_Dataset	99,441	customer_id	99,441	0	customer_unique_id repeats by design.
Products_Dataset	32,341	product_id	32,341	0	
Sellers_Dataset	3,095	seller_id	3,095	0	
Order_Payments_Dataset	103,886	order_id	99,440	4,446	Duplicates expected — needs pre-aggregation.
Order_Reviews_Dataset	99,224	order_id	98,673	551	Some orders reviewed twice.
Geolocation_Dataset	19,015	zip_code_prefix	19,015	0	Many pings per zip — must collapse.

2. Blank / missing values

Figures below reflect the state after cleaning.

Table	Column	Blanks	Treatment	Original count (pre-clean)
Orders_Dataset	order_approved_at	160	Leave blank — order never approved (genuine missing event)	160
Orders_Dataset	order_delivered_carrier_date	1,783	Leave blank — never shipped	1,783
Orders_Dataset	order_delivered_customer_date	2,965	Leave blank — never delivered	2,965
Products_Dataset	product_category_name	0	DROP these rows (user decision)	610
Products_Dataset	product_weight_g	1	Leave blank — unknown dimensions	2
Order_Reviews_Dataset	review_comment_title	87,656	Leave blank — optional free text	87,656
Order_Reviews_Dataset	review_comment_message	58,247	Leave blank — optional free text	58,247
Order_Items_Dataset_FactTable	price	0	Must be zero	0
Orders_Dataset	order_id present in payments?	1	Orders with no payment row	1

3. Data type checks

Count of cells stored as NUMBER in ID/text columns, or as non-numeric/non-date in numeric/date columns — every check should read 0.

Check	Result
order_items[order_id]	0
order_items[product_id]	0
order_items[seller_id]	0
orders[order_id]	0
orders[customer_id]	0
customers[customer_id]	0
products[product_id]	0
sellers[seller_id]	0
reviews[review_id]	0
payments[order_id]	0

Check	Result
order_items[shipping_limit_date] non-dates	0
orders[order_purchase_timestamp] non-dates	0
order_items[price] non-numeric	0
order_items[freight_value] non-numeric	0
payments[payment_value] non-numeric	0

4. Actions taken

Step	Action
2 — Data types	No change needed. All timestamps already true datetimes; price / freight_value / payment_value numeric; all ID columns text. Verified in section 3.
2 — Zip codes	Left as numbers. Customer / seller / geolocation zip prefixes are all stored numerically, so leading zeros were already lost on import (01037 → 1037). Consistent across all three tables, so joins still match.
3 — Missing events	order_approved_at (160), order_delivered_carrier_date (1,783), order_delivered_customer_date (2,965) left blank — genuine missing events, not zeros.
3 — Blank categories	610 product rows with blank product_category_name deleted. Products_Dataset: 32,951 → 32,341 rows.
4 — Geolocation	Collapsed 1,000,163 rows → 19,015 (one per zip prefix). Lat/lng = arithmetic mean of all pings; city/state = most frequent value per zip. Table resized to A1:E19016.
5 — Payments	Raw table kept (preserves payment_type detail). New sheet 'payments_by_order' / table Payments_By_Order: 99,440 rows, one per order_id. Relate this to orders, not the raw payments table.
5 — Reviews	Raw table kept. New sheet 'reviews_by_order' / table Reviews_By_Order: 98,673 rows, one per order_id (551 orders had 2+ reviews).
6 — Translation	Done (by user). Translation sheet restored: 71 rows, table 'Prodcut_Category_Name_Translation', headers product_category_name / english_product_category_name. Gap: products contains 73 distinct categories, so 2 have no English translation (see section 7). They will show blank on the English side of the relationship.
7/8 — Power Pivot	Done (by user). All tables added to the Data Model and relationships created with no errors reported. Reference steps retained in section 6 of this log.

5. Tie-out checks

All rows below should read OK.

Check	Value A	Value B	Status	Note
Payments: aggregated total = raw total	16,008,872.12	16,008,872.12	**OK**	
Payments_By_Order: one row per order_id	99,440	99,440	**OK**	
Reviews_By_Order: one row per order_id	98,673	98,673	**OK**	
Geolocation: one row per zip prefix	19,015	19,015	**OK**	
Products: no blank categories remain	0	32,341	**OK**	
Products: product_id still unique	32,341	32,341	**OK**	
Customer zips found in geolocation	14,994	14,837	**ORPHAN ZIPS**	157 customer zip prefixes have no geolocation match (known Olist gap). Relationship still valid — Power Pivot will show them under a blank member.

6. Power Pivot setup — manual steps

These steps cannot be automated from the add-in and were carried out directly in Excel.

Step	Instruction
7a	Add the product_category_name_translation sheet first, and format it as a Table (Ctrl+T) named Category_Translation.

Step	Instruction
7b	For each table: click inside it, then Power Pivot > Add to Data Model. Tables: Order_Items_Dataset_FactTable, Orders_Dataset, Customer_Dataset, Products_Dataset, Sellers_Dataset, Geolocation_Dataset, Payments_By_Order, Reviews_By_Order, Category_Translation.
7c	Do NOT add the raw Order_Payments_Dataset or Order_Reviews_Dataset to the model — they have multiple rows per order_id and will break the relationships. Use the pre-aggregated tables instead.
8a	In Power Pivot > Diagram View, create these relationships (drag FROM the many side TO the one side): • Order_Items_Dataset_FactTable[order_id] → Orders_Dataset[order_id] • Order_Items_Dataset_FactTable[product_id] → Products_Dataset[product_id] • Order_Items_Dataset_FactTable[seller_id] → Sellers_Dataset[seller_id] • Orders_Dataset[customer_id] → Customer_Dataset[customer_id] • Payments_By_Order[order_id] → Orders_Dataset[order_id] • Reviews_By_Order[order_id] → Orders_Dataset[order_id] • Customer_Dataset[customer_zip_code_prefix] → Geolocation_Dataset[geolocation_zip_code_prefix] • Products_Dataset[product_category_name] → Category_Translation[product_category_name]
8b	Sellers_Dataset[seller_zip_code_prefix] → Geolocation_Dataset: Power Pivot allows only one active relationship per table pair. Customers already uses it, so this one will be created inactive. Either leave it inactive and use USERELATIONSHIP() in measures, or duplicate Geolocation as a second table (Seller_Geolocation) for a separate active link.
8c	If Power Pivot rejects any relationship as invalid, the 'one' side still has duplicate keys — re-check section 1 of this log.

7. Open items / known gaps

Item	Detail	Impact
Untranslated categories — closed	pc_gamer | portateis_cozinha_e_preparadores_de_alimentos	Fixed: 2 rows added to Product_Category_Name_Translation (now 73 rows). English values supplied by Claude, not from the original Olist file — 'pc_gamer' kept as-is, 'portable_kitchen_food_preparers' is a translation. Coverage re-verified: 73 product categories, 73 translations, 0 missing, 0 unused. Requires a data model refresh to propagate.
Orphan customer zips — accepted	157	Customer zip prefixes with no geolocation row. Relationship is valid; these roll up under a blank member in any geography visual.
Order with no payment — accepted	1	One order_id has no row in payments. Payment measures will be blank for it.
Dropped products — accepted	610	1,603 of 112,650 order_items rows (1.42%) reference a dropped product and will fall under a blank product in the model. Consequence of the 'drop blank categories' decision. Revenue for these rows still counts in totals; it just cannot be split by category.
Table name typo — closed	Product_Category_Name_Translation	Corrected by user.

8. Calculated columns added to Orders_Dataset

Live formulas, columns I:Q.

Check	Result	Note
OnTimeFlag = On time	88,649	Delivered on/before order_estimated_delivery_date
OnTimeFlag = Late	7,827	
OnTimeFlag = Not delivered	2,965	Should equal the 2,965 blank delivery dates
On-time % (of delivered)	91.9%	Excludes never-delivered orders
Average delivery days	12.6	
Average delay when late (days)	9.6	Positive = days past the estimate

Check	Result	Note
Bucket rows total = table rows	99,441	OK
Distinct delivery buckets	5	Expect 5: 0-3, 4-7, 8-14, 15+, Not delivered

9. Lookup columns folded onto Orders_Dataset

Columns R:U.

Reason: Power Pivot propagates filters one-side → many-side only. Payments_By_Order and Reviews_By_Order sit beside Orders (1:1), not above it, so slicing by payment type or review score would not have filtered item-level revenue — every slice would return the same total. Folding these attributes onto Orders (which is the one-side of order_items) makes them propagate correctly.

Consequence: Payments_By_Order and Reviews_By_Order are now optional in the Data Model. You can remove them and their relationships, or leave them — the pivots will use the Orders columns.

Check	Result	Note
Payment value tie-out: Orders col S = raw payments	16,008,872.12	OK
Orders with no payment record	1	Expect 1
Orders with no review	768	Expect 768 (99,441 orders − 98,673 reviewed)
Distinct payment types	6	Includes the 'No payment record' bucket
Average review score (order level)	4.09	